ROLL NO: | | | | | | | | | | |

(To fill in the roll no is compulsory)

INVERTIS UNIVERSITY BAREILLY

EVEN SEMESTER EXAMINATION 2024-25
B.Tech. (CSE)/B.TECH+ M.Tech. (CSE)
Semester: VIII

SUBJECT CODE: BCS085

BIG DATA ANALYSIS

Time: 03:00 Hrs. **Maximum Marks: 50**

NOTE: THE QUESTION PAPER CONTAINS THREE SECTIONS. ATTEMPT ALL THE SECTIONS.

SECTION – A

NOTE: Attempt all parts of the following question. (10 Marks)

Q. 1. Define the following terms: (1 × 5 = 5 Marks)

(i) Big Data Analytics
(ii) Hive Metastore
(iii) MapReduce
(iv) Data Locality
(v) Map Task

Q. 2. Answer the following questions: (1 × 5 = 5 Marks)

(i) List the key characteristics of a Distributed File System.
(ii) Give one reason why Big Data is considered important.
(iii) What are the two primary phases of a MapReduce job?
(iv) What is the role of the NameNode in HDFS?
(v) You have a 200 MB file. How would HDFS store this file, assuming the default block size?

SECTION – B

NOTE:Attempt all questions. (5×2×2 = 20 Marks)

Q.3.(i)(a) Explain what each of the "Four Vs" represents in the context of Big Data.

OR

(b) Explain in your own words how a Distributed File System differs from a traditional local file system.

(ii)(a) Evaluate which of the "Four Vs" you believe is the most significant challenge when dealing with Big Data and why.

OR

(b) Analyze the trade-offs between data consistency and availability in the design of a Distributed File System.

Q.4.(i) (a) Explain the difference between predictive and prescriptive analytics in Big Data.

OR

(b) Explain the role of the Map function in MapReduce. What are its typical inputs and outputs?

(ii)(a) Can MapReduce be used for matrix-vector multiplication? Explain the role of HDFS in the Hadoop ecosystem.

OR

(b) What are the two primary components of the Hadoop core architecture? Discuss.

SECTION – C

NOTE: Attempt all questions. (10 × 2 = 20 Marks)

Q.5.(a) What is the default block size in HDFS? Analyze the impact of increasing the block size in HDFS with a suitable example.

OR

(b) Explain the difference between hadoop fs -put and hadoop fs -copyFromLocal. You want to create a directory named "data" in HDFS. What command would you use?

Q.6.(a) Explain how different Hadoop ecosystem components work together to solve a Big Data problem. Provide a simple example.

OR

(b) Describe the role of ZooKeeper in the Hadoop ecosystem. You have unstructured data in HDFS and must perform complex data transformations and analysis using a procedural data flow language. Which Hadoop ecosystem component would you likely use?